decision trees using suitable Python tools. Finally, students will interpret analytical results and present meaningful insights through well-organised outputs, visualisations, and brief reports, demonstrating practical skills required for real-world data analysis tasks.

TOPICS TO BE COVERED

Module	Module Title	Topics to be Covered (Practical)
1	Python Fundamentals for Data Science	Variables, data types, lists, tuples, dictionaries, loops, functions
2	NumPy for Numerical Operations	Array creation, indexing, slicing, mathematical operations on arrays
3	Pandas for Data Handling	Series and DataFrame, importing CSV files, data selection, filtering
4	Data Cleaning and Preparation	Handling missing values, removing duplicates, data transformation
5	Exploratory Data Analysis and Visualization	Descriptive statistics, summary measures, line chart, bar chart, histogram, pie chart using Matplotlib
6	Statistical Measures and Probability using Python	Mean, median, mode, standard deviation, variance, simple probability examples
7	Basic Data Science Techniques	Implementation of classification, decision tree, and clustering on simple datasets; mini case study

SUGGESTED READING:

1. Data Science and Business Analytics — S. P. Rajagopalan, Publisher: Wiley India
2. Statistics for Data Scientists: 50 Essential Concepts — Peter Bruce & Andrew Bruce Publisher: O'Reilly (Indian edition available)
3. Fundamentals of Statistics — S. C. Gupta & V. K. Kapoor, Publisher: Sultan Chand & Sons
4. Data Science: Introductory and Advanced Topics — N. G. Das, Publisher: McGraw Hill Education India
5. Machine Learning & Data Science – An Introduction to Statistical Learning Methods with R and Python — Daniel D. Gutierrez, Publisher: CRC Press (Indian edition)
6. Python for Data Science — Dr. Basant Agarwal & Dr. Sumit Chauhan, Publisher: Oxford University Press India

SUBJECT NAME: Introduction to Machine Learning **Credit: 5 (3L + 4P)**
SUBJECT CODE: BCAC701B

COURSE OBJECTIVE:

This course aims to introduce students to the fundamental concepts and principles of machine learning for data-driven problem solving. It provides an understanding of how computers learn from data using supervised and unsupervised learning techniques. Students will explore essential ideas such as data preprocessing, feature selection, model training, testing, and evaluation. The course covers basic algorithms including regression, classification, clustering, and decision trees with simple examples. Emphasis is given to understanding the workflow of building machine

learning models and interpreting their results. Learners will also become familiar with practical tools and libraries used to implement machine learning solutions. By the end, students will gain the foundational knowledge required to apply machine learning methods to real-world datasets and analytical tasks.

Course Outcome	
CO1	Explain the basic concepts, types, and applications of machine learning in data-driven problem solving.
CO2	Prepare datasets through preprocessing, feature selection, and data transformation for model building.
CO3	Apply fundamental machine learning algorithms for regression, classification, and clustering tasks.
CO4	Evaluate and interpret the performance of machine learning models using appropriate metrics.
C05	Use suitable tools and libraries to implement simple machine learning solutions on real datasets.

DETAILED SYLLABUS:

Module	Module Title	Detailed Topics Covered	Hours	Marks
1	Introduction to ML & Data Preprocessing	Meaning and scope of ML; types of learning; applications; ML workflow; data types; preprocessing; handling missing values; feature selection; training and testing data	9	20
2	Regression Techniques	Simple linear regression; multiple regression (basic idea); model prediction; evaluation using MAE, MSE, RMSE	9	20
3	Classification Techniques	Logistic Regression; KNN; Naïve Bayes; confusion matrix; accuracy, precision, recall	9	20
4	Clustering and Decision Tree	K-Means clustering; centroid concept; decision tree structure; entropy and information gain; tree construction	9	20
5	Model Evaluation and Applications	Model validation; overfitting and underfitting; cross-validation (basic idea); ML applications; case study	9	20
		TOTAL	45	100